

OpenClaw OLED Display Tutorial

Before starting, please complete [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [04-AI Voice Interaction Configuration], and [05-Three Dialogue Configuration Tests] in the `Preparation Before Use` directory. If you are only using TUI/WhatsApp, you can skip `04`. This article assumes the basic environment is ready and only covers OLED-related content.

1. Tutorial Objectives

This article demonstrates OpenClaw's natural language control capability through OLED peripherals. After completing this tutorial, readers will be able to:

- Understand the basic approach to controlling OLED display with OpenClaw
- Communicate with OpenClaw via WhatsApp, TUI, and voice commands
- Complete four OLED cases: text display, metric display, weather card, animation display
- Understand the complete chain from "user input" to "screen display"

2. Hardware Preparation

- OpenClaw main control board
- OLED display screen [default SSD1306 128x32 / I2C address 0x3C]
- [Optional] Microphone module
- Wiring diagram



3. Software Preparation

Code entry points used in this tutorial:

- Low-level driver code: `/home/pi/hardware_ha1_demo/`
- OLED control command: `/home/pi/hardware_ha1_demo/oled`
- Temperature and humidity display with OLED: `/home/pi/hardware_ha1_demo/dht_oled`
- Voice entry: `/home/pi/voice_to_openclaw/src/main.py`

The `oled` tool already supports the following capabilities:

- Display text `text`
- Display image `image`
- Display animation `animation`
- Display weather card `weather`
- Clear screen `clear`

If you just want to quickly display temperature and humidity readings on OLED, you can also directly use `dht_oled`.

```
dht_oled #Display temperature and humidity data on OLED
oled text --text 'Hi, world' #Display text
oled weather --location 'Shenzhen' --condition 'Sunny' --temperature '26°C' --
humidity '57%' #Weather card
oled image --path ~/hardware_hal_demo/assets/oled/mountain_photo_128x32.png
#Display image
oled animation --path ~/hardware_hal_demo/assets/oled/sunrise_anim_128x32.gif
#Display GIF animation
oled clear # Clear screen
```

Note: The GIF and image display can be generated by AI with 128x32 resolution, then placed in this path: `~/hardware_hal_demo/assets/oled/`

For more commands, see the md document in this folder:

```
/home/pi/hardware_hal_demo/README.md
```

If the low-level commands execute normally, it means the hardware and Raspberry Pi board are properly connected with normal communication.

4. Testing OLED Display

Before starting this tutorial, please complete an `openclaw tui` basic conversation self-check first; if you haven't done this yet, see [05-Three Dialogue Configuration Tests]. After confirming OpenClaw can respond normally, proceed with the OLED-specific test.

Enter TUI by typing the following in the terminal:

```
openclaw tui
```

```
pi@raspberrypi: ~  
File Edit Tabs Help  
pi@raspberrypi:~ $ openclaw tui  
[!!!]OpenClaw 2026.3.12 (6472949)  
Your terminal just grew claws-type something and let the bot pinch the busywork.  
  
openclaw tui - ws://127.0.0.1:18789 - agent main - session main  
connecting | idle  
agent main | session main | unknown | tokens ?  
  
18:28:21 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members  
18:28:21 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_u  
ser_get_messages, feishu_im_user_get_thread_messages, feishu_im_user_search_messages  
18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki  
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media  
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node  
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet  
18:28:21 [plugins] feishu_im: Registered feishu_im_bot_image  
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)  
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc  
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool  
  
session agent:main:main  
gateway connected | idle  
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 0/1.0m (0%)  
  
|
```

You can try the following in the terminal:

- Display "Hi, OpenClaw" on OLED
- Display "Today we start learning OLED" on the screen
- Clear the OLED screen
- Display current status "System OK" on OLED

```
pi@raspberrypi: ~  
File Edit Tabs Help  
Display "Hi, OpenClaw" on OLED  
  
"Hi, OpenClaw" is now displayed on the OLED screen. ✓  
  
Display "Today we start learning OLED" on the screen  
  
"Today we start learning OLED" is now displayed on the screen. The text was  
automatically wrapped into two lines to fit the OLED display. ✓  
  
Clear the OLED screen  
  
OLED screen cleared. ✓  
connected | idle  
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens  
46k/1.0m (5%)  
  
|
```

5. Three Ways to Communicate with OpenClaw

Method	Suitable Scenario	Advantages	Suggested Use
WhatsApp	Remote control, classroom demonstration	Easy to connect, suitable for multi-person experience	Demonstrate "remote natural language control"
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	First to get the control link working
Voice Code	Voice interaction, complete AI experience	Closest to real conversation experience	As the final demonstration solution

If you have the optional AI voice module + speaker, the TUI and WhatsApp conversations need to broadcast OpenClaw's replies. You can run the following program in the terminal. This is not needed for AI voice module conversations.

- Modify the TTS enable parameter

```
/home/pi/voice_to_openclaw/.env
```

```
ENABLE_TTS=true # This parameter controls whether to broadcast replies. After saving, run the program below in the terminal.
```

- Terminal input

```
cd ~/voice_to_openclaw
source .env/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) pi@raspberrypi:~/voice_to_openclaw $ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/pi/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 WhatsApp Solution

For WhatsApp integration steps, please refer directly to [05-Three Dialogue Configuration Tests]. This section only includes dialogue examples suitable for the OLED tutorial.

After creating, you can converse directly, for example:

- Display "Welcome to OpenClaw" on OLED
- Help me display current temperature and humidity data on OLED
- Clear the OLED screen

19:52



< 2



+86 153 3885 7526 (你)

给自己发消息

with a nice blue glow.

19:44 ✓✓

Display "welcome to openclaw" on OLED

19:46 ✓✓

Done! "welcome to openclaw" is now displayed on the OLED screen. The text was split across two lines due to the screen's 128-character width limitation.

19:47 ✓✓

Display "welcome to openclaw" on OLED

19:52 ✓✓

Done! "welcome to openclaw" is now displayed on the OLED screen. The text was split across two lines due to the 128-character width limit of the display.

19:52 ✓✓



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space

return



5.2 TUI Solution

If you have already completed the TUI self-check from chapter 0, simply type the following command in the terminal to continue the conversation:

```
openclaw tui
```

```
pi@raspberrypi:~$ openclaw tui
[Hi] OpenClaw 2026.3.12 (6472949)
  Your terminal just grew claws—type something and let the bot pinch the busywork.

openclaw tui - ws://127.0.0.1:18789 - agent main - session main
connecting | idle
agent main | session main | unknown | tokens ?

18:28:21 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members
18:28:21 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_user_get_messages, feishu_im_user_get_thread_messages, feishu_im_user_search_messages
18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet
18:28:21 [plugins] feishu_im: Registered feishu_im_bot_image
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | baillian/qwen3.5-plus | tokens 0/1.0m (0%)
```

5.3 Voice Code Solution

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up flow, please refer directly to [04-AI Voice Interaction Configuration]. After waking up, you can directly speak control content, for example:

- Display "Hi" on OLED
- Change the screen content to "Welcome to OpenClaw"
- Clear the screen

```
pi@raspberrypi: ~/voice_to_openclaw
File Edit Tabs Help
2026-05-14 17:25:01 [INFO] voice_to_openclaw:run:26 - voice_to_openclaw starting
[系统] voice_to_openclaw 已启动
[系统] ASR=xunfei_ws/iat/en_us | 发布=gateway_chat | TTS=开/xunfei_ws | 唤醒=串口
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 8520 ms 语音, 正在转写

[你] Display high on old.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Done! "high" is now displayed on the OLED screen. 🟢
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音, 正在转写

[你] Clear the screen.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Done! The OLED screen has been cleared. 🟢
```

6. Comprehensive Cases

Note: All demonstrations below use the `openclaw tui` method for conversation. You can also choose WhatsApp or voice to complete.

6.1 Case 1: Text Display

Experiment Objective

Have OLED display a fixed text for the most basic screen control.

Effect Description

After OpenClaw receives the command, it will render the specified text on the OLED screen. This case is suitable for verifying whether text rendering, line wrapping, and screen refresh are normal.

Example Commands

- Display "Hi, OpenClaw" on OLED
- Display "Today's task is testing OLED"
- Change the screen content to "System Ready"

6.2 Case 2: Metric Display

Experiment Objective

Have OLED display single-line status metrics in a more compact way.

Effect Description

This mode is suitable for compressing short information like temperature, humidity, battery level, and status values into one line, more like a simple dashboard reading.

Example Commands

- Display "Temp 26C Hum 57%" on OLED
- Write the current status as a short message in one line
- Change the screen content to "CPU OK RAM OK"

6.3 Case 3: Weather Card

Experiment Objective

Have OLED display a more complete weather information card.

Effect Description

The screen can display location, weather, temperature, and humidity information. This case is suitable for weather stations, environmental monitoring, or information panels.

Example Commands

- Display Shenzhen sunny 26 degrees humidity 57% on OLED
- Make a simple weather card
- Format current temperature and humidity as a card-style display

6.4 Case 4: Animation Display

Experiment Objective

Have OLED play images or animations, showcasing richer visual effects.

Effect Description

Besides static text, OLED can also display bitmaps, GIF animations, or simple self-check animations. This case is very suitable for boot pages, emoji animations, or classroom demonstrations (you can prepare photos or GIF animations to display and place them in the path `/home/pi/hardware_hal_demo/assets/oled`).

Example Commands

- Play a simple animation on OLED
- Display an icon image
- Make a boot welcome animation